

Physical Science Part 6

1. According to the effects of length contraction, from the viewpoint of an observer stationary with respect to a body moving at a uniform speed relative to the observer, which one is CORRECT?

- A. The body contracts along the direction of the motion.
- B. The time it takes for a clock incorporated in the body to tick contracts.
- C. The body is not now contracted but would contract if it were accelerate.
- D. The body contracts in some direction traverse to the direction of motion.

2. April and May are identical twins. One day, April embarked on a high-speed space travel into a distance star. When he gets back home, who is older between the two?

- A. April
- B. May
- C. None of them because they are still of the same age
- D. Cannot be determined

3. An accelerated frame is a non-inertial frame. Is this TRUE?

- A. True
- B. False
- C. Sometimes true, sometimes false
- D. Not enough information given

4. What is the rest energy of a single proton which has a rest mass of 1.67×10^{-27} kg?

- A. 1.50×10 J
- B. 1.50×10^1 J
- C. 1.50×10^{-10} J
- D. 1.50×10^{-10} J

5. Which of the following statements is TRUE about the speed of light?

- A. The speed of light has a value that depends on the observer's frame of reference.
- B. The speed of light has the same value for observers in all reference frames.
- C. The speed of light is constant in an inertial frame of reference.
- D. The speed of light is slower in a moving frame of reference.

6. A person is riding a moped that is travelling at 20 m/s. What is the speed of the ball if the moped rider throws a ball forward at 4 m/s while riding the moped?

- A. 16 m/s
- B. 20 m/s
- C. 24 m/s
- D. 80 m/s

7. According to Einstein, what is considered the fourth dimension?

- A. Curled dimension
- B. Horizontal dimension
- C. Space dimension
- D. Time dimension

8. As you observe a fish inside an aquarium, the fish appears to be bigger when underwater. When observed through the side, the fish seems closer than if you observe over the top. When you look through the corner, it appears to be that there are two fishes. What explains this observation?

- A. Diffraction
- B. Interference
- C. Reflection
- D. Refraction

9. What is the phenomenon by which the incident light falling on a surface is back into the same medium?

- A. Absorption
- B. Polarization
- C. Reflection
- D. Refraction

10. Which property is responsible for alternating light and dark bands when light passes through two or more narrow slits?

- A. Diffraction
- B. Interference
- C. Polarization
- D. Refraction

11. What happens when a light wave enters into a medium of different optical density?

- A. Its speed and frequency change.
- B. Its speed and wavelength change.
- C. Its frequency and wavelength change.
- D. Its speed, frequency and wavelength change.

12. If carbon tetrachloride has an index of refraction of 1.461, what is the speed of light through this liquid? ($c = 3.00 \times 10^8$ m/s)

- A. 1.46×10^8 m/s
- B. 2.05×10^8 m/s
- C. 4.38×10^8 m/s
- D. 4.46×10^8 m/s

13. A ray of light in air is incident on an air-to-glass boundary at an angle of 36 degrees with the normal. If the index of refraction of the glass is 1.65, what is the angle of the refracted ray within the glass with respect to the normal?

- A. 18 degrees
- B. 30 degrees
- C. 46 degree
- D. 56 degrees

14. If the critical angle for internal reflection inside a certain transparent material is found 48 degrees, what is the index of refraction of the material? (air is outside the material)

- A. 0.74
- B. 1.35
- C. 1.48
- D. 1.53

15. According to the law of reflection, from what angle is light coming after it is reflected from a mirror?

- A. The same angle as the incident ray as measured from the surface of the mirror
- B. The same angle as the incident ray as measured from the normal to the mirror
- C. Some light comes directly from the object and some passes through the mirror
- D. Cannot be determined

16. Two planed mirror are positioned perpendicular to each other. A ray of rad light incident on mirror at an angle of 55° . This ray is reflected from mirror 1 and then strikes 2 and is reflected again. The angle of reflection will be _____.

- A. 35°
- B. 45°
- C. 55°
- D. 70°

17. A point object is placed in front of a plane mirror. Which is the correct location of the image by the mirror?

- A. A
- B. B
- C. C
- D. D

18. Which of the following is the BEST description of a white light?

- A. It is a mixture of every color.
- B. It is a mixture of seven color.
- C. It is a mixture of red, blue and yellow.
- D. It is a pure color that cannot be broken up.

19. Which of the following observation is NOT true whenever a system is in chemical equilibrium?

- A. Attainment of equilibrium requires time.
- B. The rates of opposing reactions are equal.
- C. The products do not change back to reactants.
- D. Concentration remain constant at equilibrium even though the opposing reactions are continuing.

20. Sunsets often appears to have a more reddish-orange color than noon. Which of the following phenomenon is responsible for this effect?

- A. Rayleigh Scattering
- B. Dispersion
- C. Polarization
- D. Refraction

21. The polarization behavior of light is BEST explained by considering light to be _____.

- A. particles
- B. transverse waves
- C. longitudinal waves
- D. unpolarized

22. Which type of light is vibrates in a variety of planes?

- A. Electromagnetic
- B. Polarized
- C. Transverse
- D. Unpolarized

23. Light is passed through a filter whose transmission axis is aligned horizontally. It then passes through a second filter whose transmission axis is aligned vertically. What will happen to the light after passing through both filters?

- A. It will be polarized
- B. It will be un polarized
- C. It will be entirely blocked.
- D. It will return to its original state.

24. Light rays enter a material and scatter. Which BEST describes what happens to the light rays.

- A. Bounce straight back
- B. Move straight through
- C. Spread out in all direction
- D. Move through but in a new direction

25. The energy generated by the suns travels to Earth as electromagnetic waves of varying lengths. Which statement correctly describes an electromagnetic wave with a long wavelength?

- A. It has a high frequency and low energy.
- B. It has a high frequency and high energy.
- C. It has a low frequency and can travel through a vacuum.
- D. It has a low frequency and needs a medium to travel through.

26. Why is the sky blue?

- A. It reflects the color of the ocean
- B. The color of the outer space is blue.

- C. Blue light is more readily scattered than red light by nitrogen.
- D. Red light is scattered more easily than blue light by dust particles.

27. What human activity has added the most carbon in the atmosphere?

- A. Burning fossils fuels
- B. Mining fossil fuel
- C. Increasing soil erosion
- D. Cutting down the rainforests

28. Which of the following BEST describes the image for a thin diverging lens that forms whenever the magnitude of the object distance is less than that of the lens' focal length?

- A. Inverted, reduced, and real
- B. Inverted, enlarged, and real
- C. Upright, reduced, and virtual
- D. Upright, enlarged and virtual

29. Acid rain is formed when sulfur oxides react with water in air. It causes lakes and streams decrease in pH which makes it unable to support fish and plant life. Which is the LEAST means of solving this problem?

- A. Limestone may be added to the water basins.
- B. Neutralize sulfur oxides with basic compounds.
- C. Reduction of sulfur emissions from the industrial establishments.
- D. Providing absorbent colloidal materials to screen acid rain before hitting the water.

30. Which of the following would decrease the rate in which a solid dissolves in a liquid?

- A. Supersaturating the solution
- B. Applying pressure into the solution
- C. Grinding the solid into smaller pieces
- D. Increasing the temperature of an exothermic reaction

31. Which factors affects the rate of dissolution when powdered sugar dissolves more rapidly than granulated sugar?

- A. Application of heat
- B. Nature of solute
- C. Particle size
- D. Stirring

32. What is the physical evidence that the core is composed mostly in iron?

- A. Volcanoes regularly erupt material from the core to the surface.
- B. Scientist have sampled the core and determined its composition.
- C. The known mass of Earth requires material of high density at the core.
- D. A and B

33. Which of the following comprised the largest portion of Earth's volume?

- A. Crust

- B. Inner core
- C. Mantle
- D. Outer core

34. Which of the following statements about the Mohorovicic discontinuity is False?

- A. The Moho separates the crust from the mantle.
- B. The Moho marks the top of a partially molten layer.
- C. The Moho separates denser rocks below from less dense rocks above.
- D. Seismic wave speeds up as they pass across the Moho heading downward.

35. Where do P waves travel fastest?

- A. Upper mantle
- B. Lower mantle
- C. Outer core
- D. Inner core

36. What is the driving force for the movement of the lithospheric plates?

- A. Heat from the sun
- B. Heat in the atmosphere
- C. Unequal distribution of heat within the Earth
- D. Unequal distribution of heat in the oceans

37. Which evidence supports Alfred Wegener's Continental Drift Theory?

- A. Major rivers on different continents match
- B. Land bridge still exist that connect major continents
- C. The same magnetic direction exist on different continents
- D. Fossils of the same organism have been found on different continents

38. Which of the following is TRUE based on the Plate Tectonics Theory?

- A. The asthenosphere is strong and rigid.
- B. The lithosphere is divided into plates.
- C. The asthenosphere is divided into plates.
- D. The asthenosphere moves over the lithosphere.

39. What type of boundary occurs where two plates move together, causing one plate to descend into the mantle beneath the outer plate?

- A. Convergent boundary
- B. Divergent boundary
- C. Transform fault boundary
- D. Transitional boundary

40. Which are considered as most destructive earthquake waves?

- A. P waves
- B. Q waves
- C. S Waves
- D. Surface waves

Answer Keys:

1. A
2. B
3. A
4. D
5. C
6. C
7. D
8. D
9. C
10. B
11. B
12. B
13. A
14. B
15. B
16. A
17. D
18. A
19. C
20. A
21. B
22. D
23. C
24. C
25. C
26. C
27. A
28. C
29. D
30. D
31. C
32. C
33. C
34. B
35. B
36. C
37. D
38. B
39. A
40. D

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